

Lab – Configure a Switch Using Console Connectivity

Topology



Objectives

Part 1: Access a Cisco Switch through the Serial Console Port

Part 2: Display and Configure Basic Device Settings

Background / Scenario

Various models of Cisco routers and switches are used in all types of networks. These devices are managed using a local console connection or a remote connection. Nearly all Cisco devices have a serial console port to which you can connect. Newer models used in this lab, such as Cisco 4221, also have a USB console port.

In this lab, you will learn how to access a Cisco device via a direct local connection to the console port, using the terminal emulation program called Putty. You will also learn how to configure the serial port settings for the Putty console connection. After you have established a console connection with the Cisco device, you can display or configure device settings. You will only display settings and configure the clock in this lab.

Required Resources

- 1 Switch (Cisco 2960 with Cisco IOS Release 15.0(2) lanbasek9 image or comparable)
- 1 PC (Windows with a terminal emulation program, such as Putty or Tera Term)
- Rollover (DB-9 to RJ-45) console cable to configure the switch or router via the RJ-45 console port

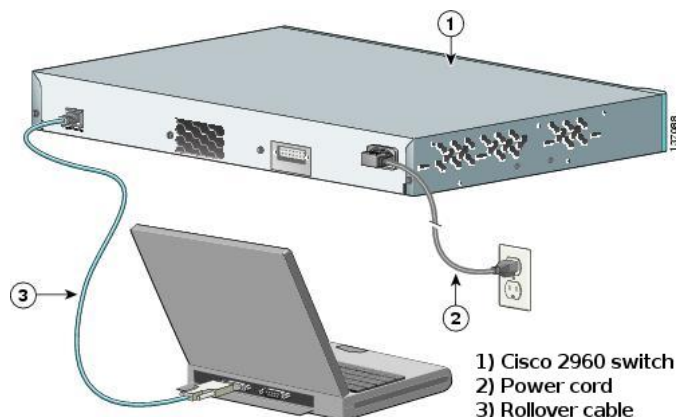
Instructions

Part 1: Access a Cisco Switch through the Serial Console Port

You will connect a PC to a Cisco switch using a rollover console cable. This connection will allow you to access the CLI and display settings or configure the switch.

Step 1: Connect a Cisco switch and computer using a rollover console cable.

- a. Connect the rollover console cable to the RJ-45 console port of the switch. Connect the other cable end to the serial COM port on the computer.



Note: Serial COM ports are no longer available on most computers. A USB-to-DB9 adapter can be used with the rollover console cable for console connection between the computer and a Cisco device. USB-to-DB9 adapters can be purchased at any computer electronics retailer.

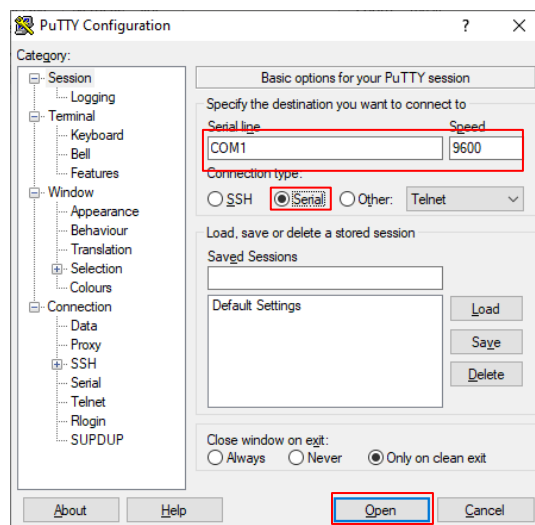
Note: If using a USB-to-DB9 adapter to connect to the COM port, you may be required to install a driver for the adapter provided by the manufacturer of your computer. To determine the COM port used by the adapter, please see Part 3 Step 3. The correct COM port number is required to connect to the Cisco IOS device using a terminal emulator in Step 2.

- b. Turn on the Cisco switch and computer.

Step 2: Configure Putty to establish a console session with the switch.

Putty is a terminal emulation program. This program allows you to access the terminal output of the switch. It also allows you to configure the switch.

- a. Start Putty from the Desktop Icon or by clicking the **Windows Start** button located in the task bar. Search for or locate **Putty** under **All Programs**.
- b. In the Connection type: click the **Serial** radio button. Verify that the COM1 port is selected and that the Speed (baud rate) is set for 9600
- c. Click **Open** to continue.
- d. When you can see the terminal output, you are ready to configure a Cisco switch.
- e. Take a screenshot of the terminal window. (You will need to upload the screenshot as part of your required submission for this exercise.



Part 2: Display and Configure Basic Device Settings

In this section, you are introduced to the user and privileged EXEC modes. You will determine the IOS version, display the clock settings, and configure the clock on the switch.

Step 1: Display the switch IOS image version.

- After the switch has completed its startup process, the following message is displayed. Enter **n** to continue.

```
Would you like to enter the initial configuration dialog? [yes/no]: n
```

Note: If you do not see the above message, please contact your instructor to reset your switch to the initial configuration.

- While you are in the user EXEC mode, display the IOS version for your switch.

```
Switch> show version
```

```
Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE, RELEASE SOFTWARE (fcl)
```

```
Technical Support: http://www.cisco.com/techsupport
```

```
Copyright (c) 1986-2012 by Cisco Systems, Inc.
```

```
Compiled Sat 28-Jul-12 00:29 by prod_rel_team
```

```
ROM: Bootstrap program is C2960 boot loader
```

```
BOOTLDR: C2960 Boot Loader (C2960-HBOOT-M) Version 12.2(53r)SEY3, RELEASE SOFTWARE
```

```
Switch uptime is 2 minutes
```

```
System returned to ROM by power-on
```

```
System image file is "flash://c2960-lanbasek9-mz.150-2.SE.bin"
```

```
<output omitted>
```

NOTE: The IOS image on your switch may be different

Step 2: Enter privileged EXEC mode.

You can access all switch commands from privileged EXEC mode. However, because many of the privileged commands configure operating parameters, privileged access should be password-protected to prevent unauthorized use.

The privileged EXEC command set includes the commands available in user EXEC mode, many additional commands, and the **configure** command through which access to the configuration modes is gained.

- Enter privileged EXEC mode by entering the enable command:

```
Switch> enable
```

```
Switch#
```

Notice that the prompt has changed to reflect privileged EXEC mode.

Step 3: Step 3: Configure and Verify Basic Switch Settings.

- Enter the global configuration mode. Give the switch a name:

```
Switch# configure terminal
```

```
Switch(config)# hostname <Team#> (ex: Team1)
```

```
Team1(config)# exit
```

```
Team1#
```

- Prevent unwanted DNS lookups

```
Team1(config)# no ip domain-lookup
```

- c. Enter local passwords. Use class as the privileged EXEC password and cisco as the password for console access:

```
Team1(config)# enable secret class
Team1 (config)# line con 0
Team1 (config-line)# password cisco
Team1 (config-line)# login
Team1 (config-line)# exit
Team1 (config)# line vty 0 15
Team1 (config-line)# password cisco
Team1 (config-line)# login
Team1 (config-line)# exit
```

- d. Configure and enable the SVI according to the Addressing Table

```
Team1 (config)# interface vlan 1
Team1 (config-if)# ip address 192.168.1.1 255.255.255.0
Team1 (config-if)# no shut
```

- e. Enter a login MOTD banner to warn about unauthorized access

```
Team1(config)# banner motd #
Enter TEXT message. End with the character '#'.
Unauthorized Access Prohibited!!!
Team1(config)# exit
```

- f. Display the current configuration. Provide a screenshot as shown below:

```
Team1# show running-config
Building configuration...

Current configuration : 1409 bytes
!
! Last configuration change at 03:49:17 UTC Mon Mar 1 1993
!
version 15.0
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname Team1
```

```
% Invalid input detected at '^' marker.

TeamMJ#show running-config
Building configuration...

Current configuration : 1511 bytes
!
! Last configuration change at 00:07:07 UTC Mon Mar 1 1993
!
version 15.0
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname TeamMJ
!
boot-start-marker
boot-end-marker
!
enable secret 5 $1$MVJH$DjAU5hinRowU8BLIMijpZ/
!
no aaa new-model
system mtu routing 1500
!
!
no ip domain-lookup
!
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
vlan internal allocation policy ascending
!
!
!
!
!
!
interface FastEthernet0/1
!
interface FastEthernet0/2
!
interface FastEthernet0/3
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
```

```
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
 ip address 192.168.1.1 255.255.255.0
!
 ip http server
 ip http secure-server
!
!
 banner motd ^C "Unauthorized Access Prohibited !!! " ^C
!
 line con 0
  password cisco
  login
 line vty 0 4
  password cisco
  login
 line vty 5 15
  password cisco
  login
!
end

TeamMJ#
```

- g. Record your answers to the reflection questions below
- h. Save this document and upload it and the screenshot on to Canvas as your submission for this lab.

Reflection Question

1. How do you prevent unauthorized personnel from accessing your Cisco device through the console port?

You can use physical security controls to block the ports and have supervision around the devices at all times.

2. What are the advantages and disadvantages of using the serial console connection compared to the USB console connection to a Cisco router or switch?

The advantages include that other malicious users cannot use their generic laptops to connect to the switch as not many modern laptops have serial console connection ports. The disadvantages include the opposite of the advantage where only specific clients can use the console port and cables to configure and manage the switches.